

Verification Report

ICCES-260 camera-ready — AI-Powered Unified Mobility and Delivery System Using Dynamic Feasibility Analysis

Experiments run · evidence used · audit findings · cross-check · limitations · sign-off items

Verification Report — ICCES-260 Camera-Ready

Paper: AI-Powered Unified Mobility and Delivery System Using Dynamic Feasibility Analysis **Authors:** Sivasubramanian M, S. R. Ramya, Rakshana S, Sadhana T **Base manuscript:** the uploaded revised camera-ready docx (edited, not regenerated from scratch) **Date of this pass:** 1 September 2026

0. ITEM FOR YOU AND YOUR ADVISOR — raised, not resolved

`docs/METRIC_DEFINITIONS.md` carries the project’s own warning, written by whoever performed the R3/R4 correction:

“Every file under `backend/evaluation/results/` still carries its original timestamp, while `framework.py`, `make_admfe_report.py`, `final_validation.py` and `analyze_unified.py` were all modified by the R3/R4 correction, and `dmfe/optimizer.py` was modified by the P0-3 fix. **The published tables therefore predate both corrections and must be regenerated before the numbers are quoted anywhere.**”

The file timestamps are consistent with that warning: every result file is dated 9 August, while `framework.py`, `make_admfe_report.py`, `final_validation.py` and `analyze_unified.py` are dated 17 August.

What this means. Tables III, IV and V, the abstract figures and the conclusion figures all trace correctly and exactly to `admfe_repetitions.json` — I verified 13 of 13 derived quantities against that file this pass, and 17 of 17 in the previous pass. What I cannot verify is whether `admfe_repetitions.json` still reflects the current engine. If the R3/R4 and P0-3 changes altered any measured quantity, the published tables are accurate to the *August 9* engine, not to the code as it stands.

This is separate from the blockers in §1 and §2 below. Those are about what could not be *added*. This one is about whether what is already there is still current. It is a decision for you and your advisor, and I have not attempted to resolve it. The project’s own consistency checker (`scripts/check_results_consistency.py`, invariant C4) is described in that document as currently failing on exactly this point.

1. What was actually run this pass

No experiments were run. None could be.

Attempted	Outcome
Regenerate the per-trip DB from the 36 seed-matched runs	Blocked
Build a simplified joint OR-Tools baseline	No-go

Root cause, verified in both environments. `app/dmfe/optimizer.py:44` imports `from ortools.constraint_solver` import `pywrapcp`, `routing_enums_pb2` at module scope, and `evaluation/framework.py:123` imports the pipeline, so OR-Tools is a hard dependency of any run. The project virtual environment is Windows-layout and ships Windows binaries only (`.venv/Lib/site-packages/ortools/.libs/ortools.dll`, `abseil_dll.dll`, `libscip.dll`, ...). Pointing Python at it from Linux gives:

```
sqlalchemy 2.0.51 -> .../backend/.venv/Lib/site-packages/sqlalchemy      # pure Python, loads
ImportError: cannot import name '_pywrapcp' from 'ortools.constraint_solver'
```

PyPI is proxy-blocked in both available environments (`403 Forbidden` from the desktop VM; `No matching distribution found for ortools` and `sqlalchemy` in the cloud container), and the project contains no wheelhouse, vendor directory or `.whl` cache.

Additionally, the original per-trip data is gone. `framework.py` recreates `evaluation/experiments/eval.db` on every run; that directory is now empty. I probed all seven JSON files in `evaluation/results/` for trip-level records — none contain any.

Not used, deliberately. `backend/dmfe_dev.db` (3,717 requests / 2,410 trips, 9 days) and `backend/qa_test.db` (1,131 / 1,364, 3 days) *do* contain joinable typed data, but they are development and QA databases accumulated from interactive use under unknown and varying engine configurations, with no seeds and no workload labels. They are not the 36 runs. Per your instruction and my own reasoning, nothing from them appears in the manuscript or the response document.

A second, deeper obstacle for per-service results. Even a successful re-run would not produce per-type efficiency figures. Service type is tagged — `SimulationRequest.request_type` (`models.py:180`) carries ride/food/parcel and `Trip.request_ids_json` links trips back to requests — but outcome metrics are computed **per trip**, and at the two larger workloads 83.3% of trips are shared, routinely carrying requests of different types. Splitting one shared trip's distance or fuel between a food order and a passenger ride needs a cost-allocation convention the system does not define. Producing one would mean inventing the assumption, not measuring it.

2. Evidence used

Evidence	Path	Used for
36-run aggregates	backend/evaluation/results/admfe_repetitions.json	Tables III–V, Figs. 2–3, all reported figures
Statistical summary	backend/evaluation/results/final_statistical_summary.csv	cross-check of the above
Closed-loop learning run	backend/evaluation/results/learning_validation.md	Section V-E
Engine configuration	evaluation/framework.py (SYSTEM_CONFIG , FLEET_SPEC)	Table II
Scoring formulas	app/dmfe/scoring.py , app/dmfe/score_engine.py	equations (1)–(7)
Driver selection	app/dmfe/driver_selection.py	equation (8), five coefficients
Routing objective	app/dmfe/optimizer.py (arc-cost callback)	equation (9)
Adaptive stack	app/dmfe/adaptive/weights.py , learning.py	equations (10)–(11), Section III-D
Dispatch order	app/dmfe/pipeline.py , driver_selection.py:767	Fig. 1, Section III-C
Data handling	app/db/models.py ; grep for anonymisation/retention/encryption returned nothing	Section VI privacy limitation
Reviewer comments	Acceptance letter.pdf (ICCES-260)	response document

3. Audit findings and what was fixed

Confirmed mismatches — all fixed:

#	Finding	Fix
1	Broken sentence in Section I-A. The manuscript read "...under three commercial labels, taxi records established that..." — the clause "Shareability analysis of city-scale" had been lost in an earlier revision, leaving a lower-case sentence fragment. Present in the uploaded base.	Clause restored
2	Section III-B ordering. The tolerance discussion was prepended to the paragraph that introduces equation (5), so it appeared between equation (4) (route similarity) and the time-window definition, discussing time tolerances before the time term existed. Confirmed from <code>word/document.xml</code> paragraph order, not from text extraction.	Split and relocated to <i>after</i> the global-W paragraph. Order is now: Pickup (3) → Route similarity (4) → Time window (5) → global W is 20 min → per-type practice → Capacity (6)
3	Table II "8 vans (13)" read as a reference to a non-existent equation (13).	Rewritten as "60 vehicles at 40 / 25 / 15.5 / 13 / 8 km/L"
4	Two contractions ("what's" ×2) in the preserved III-B passage, informal for IEEE.	De-contracted during the rewrite
5	Stray space before a full stop ("service types .") in the same passage.	Fixed
6	First-person "we" in Section IV-C, inconsistent with the paper's third-person register.	Recast in third person; substance preserved

Checked and clean — no change needed:

- No 611.3, 1,406.01, 92.1%, 98.4%, 87% anywhere.
- No Rapido, Swiggy, Delhivery, or live-provider integration claim.
- No XAI score, no DPDP/GDPR claim, no encryption or anonymisation claim. The privacy text states what is *not* implemented.
- 43-request and 2,320 appear exactly once each, only inside the Section IV-C withdrawal statement — which you asked to preserve.
- Every "trained" / "gradient descent" occurrence is a **negation or a scoping statement**; I read all five in context. None asserts execution.
- W_type vs global 20-minute: consistent. Four separate statements confirm the global constant and that no per-type value is configured or claimed.
- Figure, table, algorithm and equation references: Figs. 1–3, Tables I–V, Algorithm 1, equations (1)–(12) all referenced and all exist. No dangling references.

Reported as false positives, not changed: run-together words (Servicespecific, multiregion) are pdftotext hyphenation artifacts — the source has proper hyphens. Doubled words flagged by the scanner are Table I cell values ("No No Yes No"), not prose.

4. Cross-check results

13 of 13 derived quantities recomputed from `admfe_repetitions.json` match the paper exactly: trip reduction (36.8 / 40.4%), utilisation gain (17.0 / 17.4 pp), within-fleet CO₂ reduction (22.0 / 25.5 / 42.6 / 43.9%), mean compatibility (55.4 static, 59.2 adaptive), 398,128 pair evaluations, 199,064 distinct pairs, 1,869 trips.

Abstract, results and conclusion agree on every shared figure. The conclusion says “about 17 percentage points” where the abstract gives 17.0 and 17.4 — a deliberate rounded summary, not a mismatch.

Response document: 11 comment blocks, 11 status lines, **3 marked NOT EXPERIMENTALLY ADDRESSED** (R1.3 joint baseline, R1.4 per-service, R1.5 acceptance). Every claim the response makes about the manuscript was verified present in the manuscript. No number cited in the response is absent from the paper.

5. Formatting

- **Final page count: 8.**
- Margins verified per page: left 45.1 pt, right ≤ 567.1 , top ≥ 54.3 , bottom ≤ 719.7 — nothing overflows.
- Fonts: **Times New Roman on all 544 runs**; sizes 7 / 8 / 9 / 10 / 20 pt. No font or margin was shrunk to control length — the reduction from 9 pages back to 8 came from compressing prose in eleven passages, all listed as edits above.
- PDF exported from the final Word file, so the two match.

Page limit: unconfirmed. The ICCES-260 acceptance letter states no page limit, and I could not locate an authoritative ICCES 2026 (11th, PPG Institute of Technology) camera-ready page specification. If you supply it, I can confirm compliance or compress further. IEEE conference limits are commonly 6 or 8 pages with over-length charges above that.

6. Originality — internal check only

No external similarity or AI-detection tool was run, and no percentage from one is claimed.

What I measured locally, against your previously submitted version:

Window	Overlap (body only, references excluded)
7-gram	4.57%
8-gram	3.61%
10-gram	2.25%

Internal repetition: 21 repeated 8-grams. Ten are the four author-affiliation blocks (standard IEEE format). The rest are the five-factor list restated between the abstract and the contributions, the workload sizes, and the driver coefficients repeated between Table II and Section III-C — legitimate technical repetition of defined terms, not padding.

A true Turnitin, iThenticate or IEEE CrossCheck figure cannot be produced locally and is not supplied here.

7. Remaining limitations, as stated in the paper

Section VI carries seven, each stating what was not done: synthetic single-region workload with no live traffic; global aggregates with no per-service instrumentation; fleet-bounded throughput at $N = 250$ and 500 with no efficiency comparison there and no re-implemented joint-routing baseline; no driver or customer acceptance study; raw coordinates and addresses stored with no anonymisation, retention schedule, encryption or compliance framework; external mapping dependency with a haversine fallback and no alternative provider; weights and threshold configured rather than learned, with the softmax objective untrained.

8. Needs your sign-off before submission

1. **The METRIC_DEFINITIONS warning in §0** — whether Tables III–V should be regenerated against the current engine. This is the largest open question and it is yours to decide.
2. **Author name: “Ramaya” or “Ramya.”** The ICCES-260 acceptance record says *Ramaya*; the manuscript says *Ramya*. Unchanged by me and flagged in the response document. Confirm with Dr. V. Bindu in writing before uploading.
3. **The ICCES page limit (§5).**
4. **Author-block layout.** The reference screenshot you mentioned has not reached this session; the current 2×2 block follows the accepted manuscript. Please check it against your reference.
5. **Five references lack a DOI** — [3], [4], [5], [6], [8]. Crossref and OpenAlex both rate-limited; I would not guess. IEEE style does not require them, but add them if you want consistency.
6. **Co-author review of the Section IV-C withdrawal statement.** It states in the paper that the earlier pilot and deployment figures are withdrawn because the datasets cannot be independently reproduced. That is an accurate and defensible statement, but it is a substantive disclosure and your co-authors should see it before it is submitted.